

UMAP (Uniform Manifold Approximation and Projection) methods

Lecture 4
Course for Constructor University

Presented by

Vladimir Kukushkin

Liubov Tupikina

Outline of the course

Lecture 1 (Vladimir). PCA, MDS, Local MDS, ISOMAP

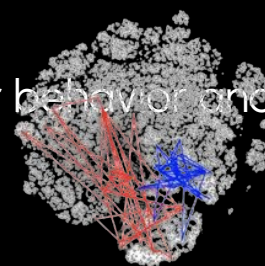
Lecture 2 (Liubov). TSNE non-linear methods

Lecture 3 (Vladimir). SVD-decomposition, collaborative filtering, RecSys

Lecture 4 (Liubov). UMAP

Lecture 5 (Liubov). *2vec, autoencoders

Lecture 6 (Vladimir). Markov model, HMM, applications in user behavior analysis

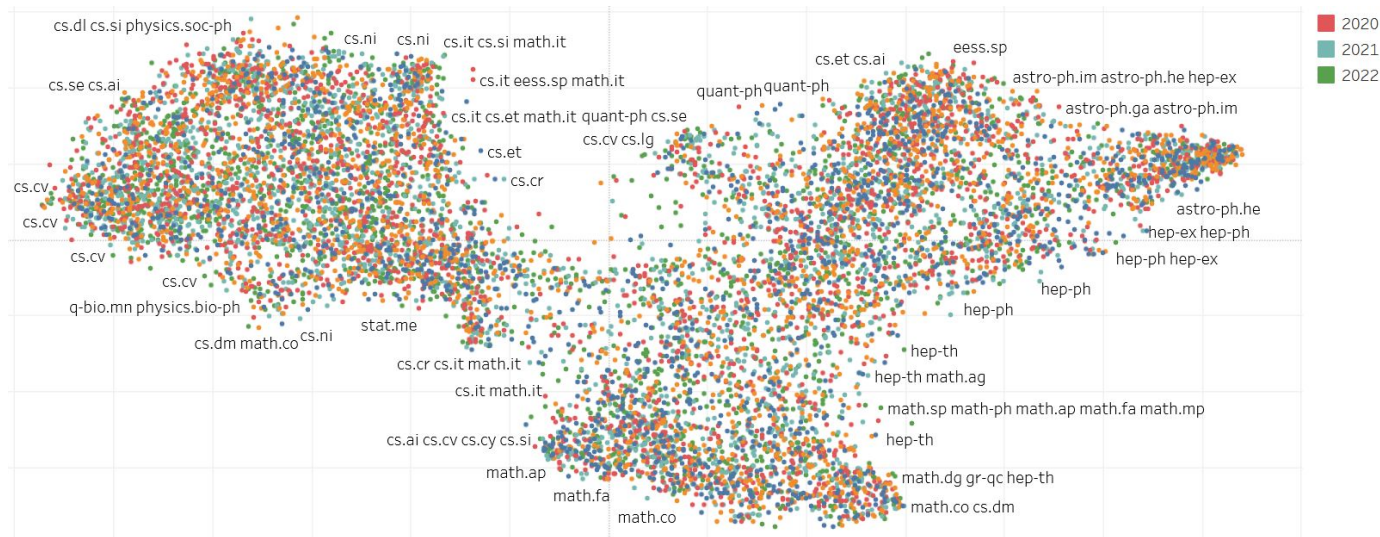


Personal projects

Datasets examples in the google [drive of the course](#)

Assignment for April 8th:

Submit 1st notebook with your personal project.



Recalling lectures 1 and 2

Why PCA, MDS , SVD or tSNE methods may not be enough for embeddings?

MDS — Idea

- ▶ Often we only care about preserving **pairwise distances**, not the absolute coordinates.
- ▶ MDS finds N points $z_1, \dots, z_N \in \mathbb{R}^q$ minimising the **stress function**:

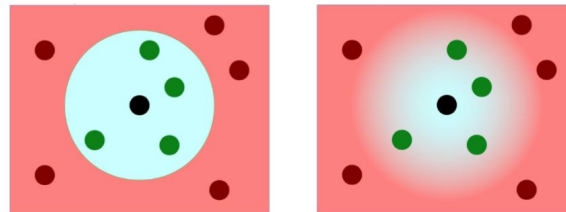
$$S_M(z_1, \dots, z_N) = \sum_{i < i'} (d_{ii'} - \|z_i - z_{i'}\|)^2 \quad (3)$$

where $d_{ii'}$ are the original pairwise distances.

- ▶ MDS is a **non-linear** method (despite a linear-algebra solution for classical MDS).

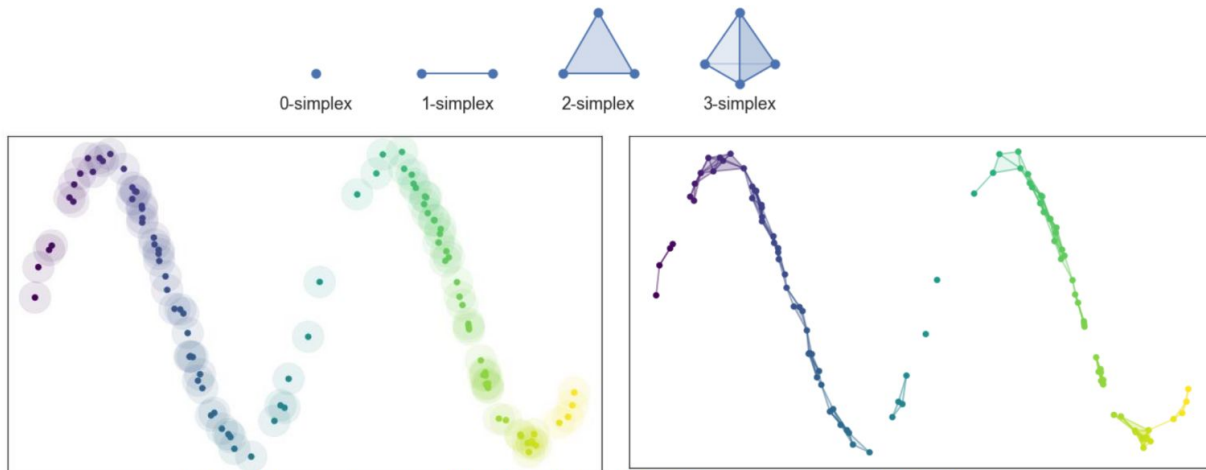
t-SNE: high-level idea

- ▶ Interpret pairwise similarities in high dimension as probabilities p_{ij} : preserving **soft neighborhood structure**.
- ▶ Find low-dimensional points with probabilities q_{ij} that match p_{ij} using cost function C : space $X \rightarrow$ space Y .
- ▶ Optimize function C by minimizing a Kullback–Leibler (KL) divergence $KL(P\|Q)$.



Uniform Manifold Approximation and Projection (UMAP)

Starts with a 'fuzzy simplicial complex' but ends with a simple weighted neighbourhood graph



Čech complex

Union of ϵ -balls around data points
— simplex exists if balls overlap.
Captures topology at scale ϵ .

Vietoris–Rips

Simplex formed when all pairwise distances $\leq \epsilon$. Computationally cheaper; approximate Čech.

Homology

Algebraic tool counting
 k -dimensional holes: H_0 =
connected components, H_1 = loops,
 H_2 = voids...

Uniform Manifold Approximation and Projection (UMAP) algorithm

1 Find k-NN

Compute k nearest neighbours for each point using approximate algorithms (e.g. NN-descent).

2 Build Fuzzy Graph

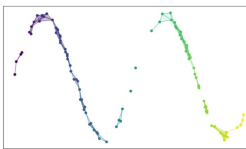
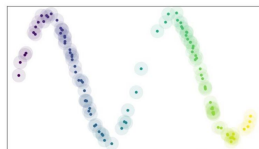
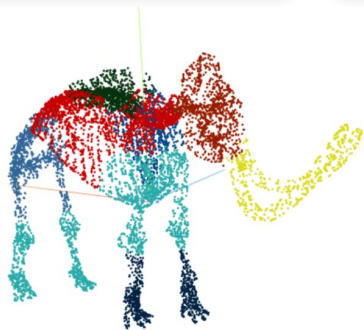
Assign membership strengths $\mu(x_i, x_j)$ using local σ_i ; symmetrise to form undirected weighted graph.

3 Initialise Layout

Embed graph nodes in $\mathbb{R}^d$ using spectral embedding (Laplacian eigenmaps) or random initialisation.

4 Optimise Layout

Minimise cross-entropy between high-D fuzzy set and low-D fuzzy set via stochastic gradient descent.



Uniform Manifold Approximation and Projection (UMAP) algorithm

Phase 1 algorithm idea: construction of the weighted neighborhood graph

For each point i , define distance to the nearest neighbour

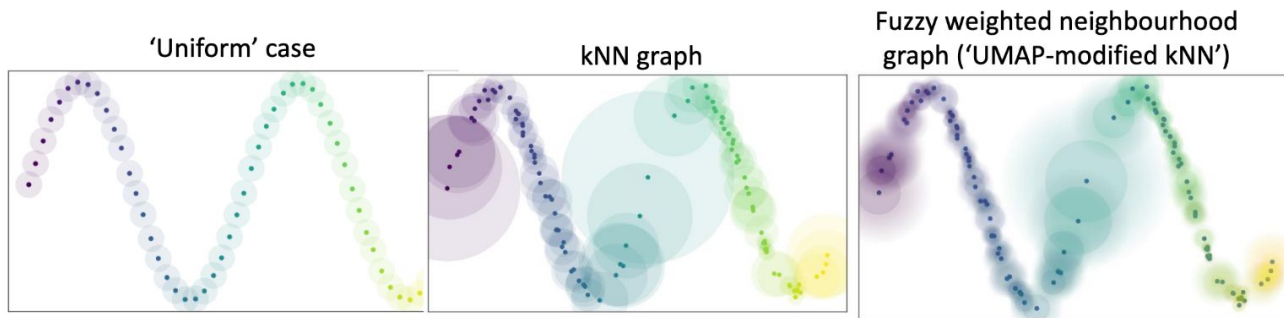
$$\rho_i = \min\{d(x_i, x_{i_j}) \mid 1 \leq j \leq k, d(x_i, x_{i_j}) > 0\},$$

Define σ_i using the following equation (local dispersion)

$$\sum_{j=1}^k \exp\left(\frac{-\max(0, d(x_i, x_{i_j}) - \rho_i)}{\sigma_i}\right) = \log_2(k)$$

Weight of the neighbourhood graph

$$w((x_i, x_{i_j})) = \exp\left(\frac{-\max(0, d(x_i, x_{i_j}) - \rho_i)}{\sigma_i}\right)$$



$k=4$

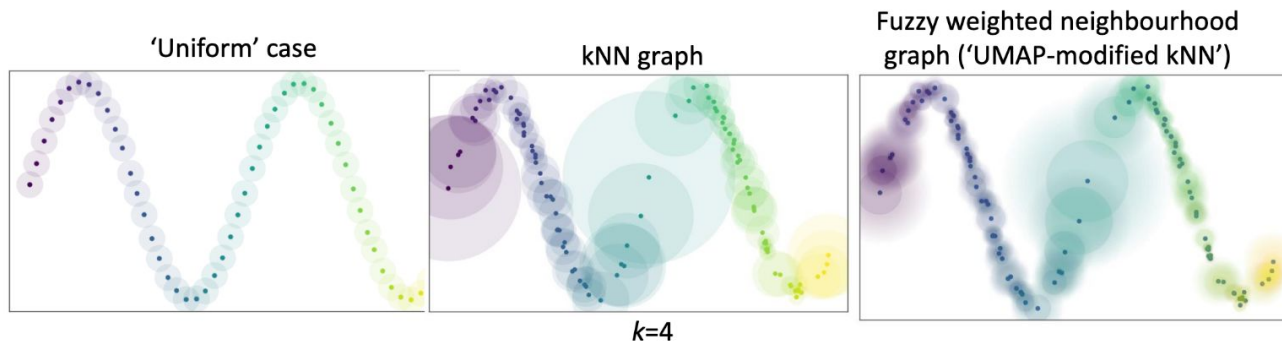
https://www.ihes.fr/~zinovyev/FundamentalsOfAI2020/lectures/Slides/Lecture4_3_tSNE_II_MAP.pdf

Victims Rips

Homology

Uniform Manifold Approximation and Projection (UMAP) algorithm

Mathematical foundations



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Uniform Manifold Approximation and Projection (UMAP) algorithm

Phase 2 algorithm idea:

Optimisation for construction of the low-dimensional graph and estimation of low-dimensional embedded graph using the cross-entropy concept:

Preservation of
small distances

Preservation of
large distances

$$\sum_{e \in E} \left[w_h(e) \log \left(\frac{w_h(e)}{w_l(e)} \right) + (1 - w_h(e)) \log \left(\frac{1 - w_h(e)}{1 - w_l(e)} \right) \right]$$

$1 \geq w_h > 0$ – weights in high-dimensional space

$1 \geq w_l > 0$ – weights in low-dimensional space

Actual algorithm is very close to force-directed
layout algorithm

UMAP method Manifold Assumption

Real-world high-dimensional data does not fill $\mathbb{R}^D$ uniformly — it lies on or near a low-dimensional manifold embedded in the ambient space.

Intrinsic dimensionality

Data has internal structure in d dimensions, even if observed in $D \gg d$.

Local linearity

Small patches of the manifold are approximately Euclidean — enabling local distance comparisons.

Geodesic distance

True distances should be measured along the manifold, not through empty ambient space.

